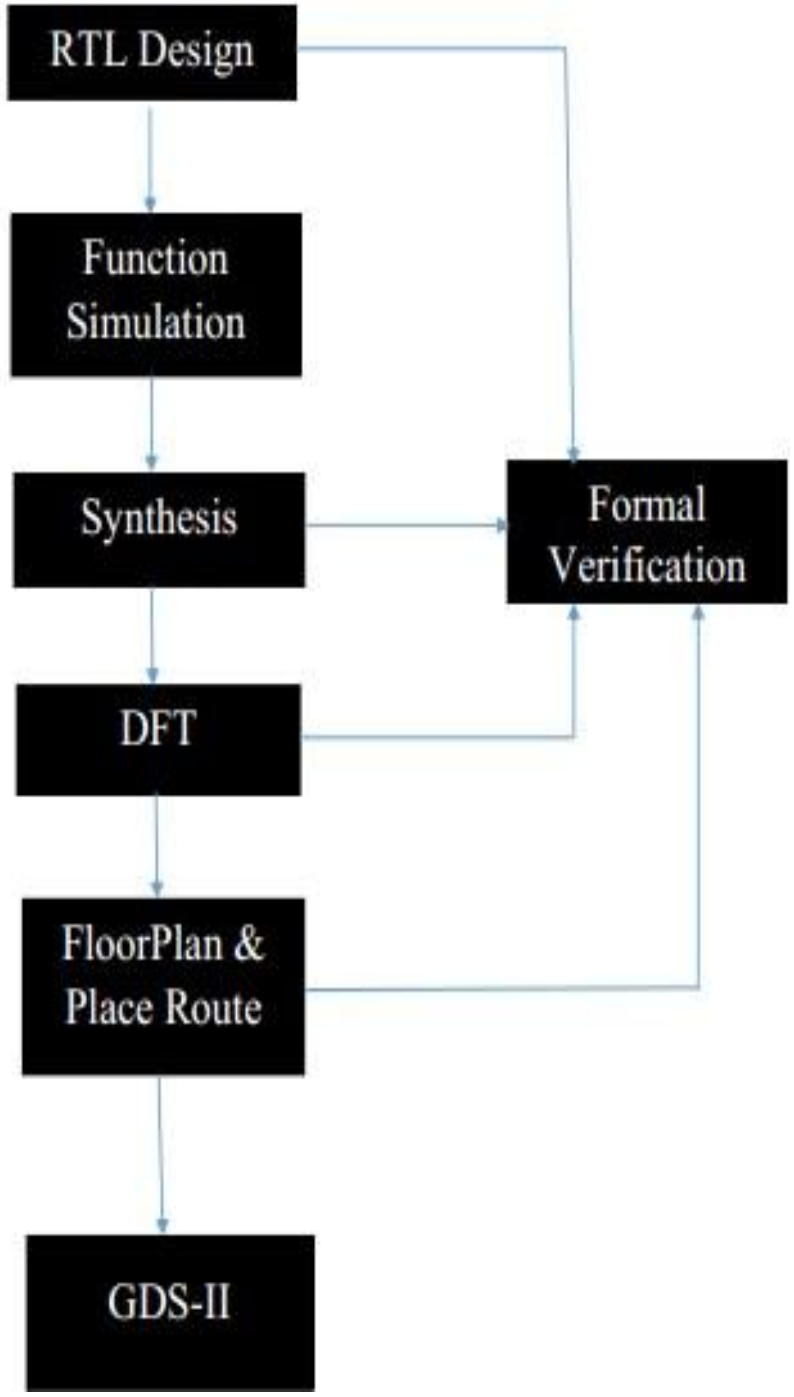
CAMBRIDGE INSTITUTE OF TECHNOLOGY

BENGALURU

RTL to GDS-II Flow

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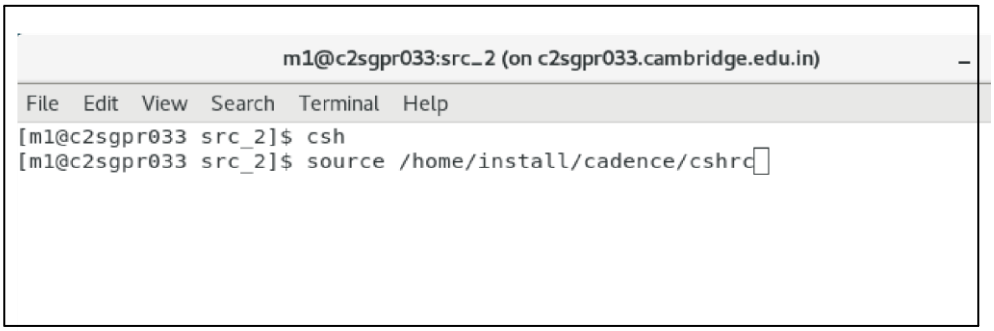
Flow Diagram of ASIC design flow :-

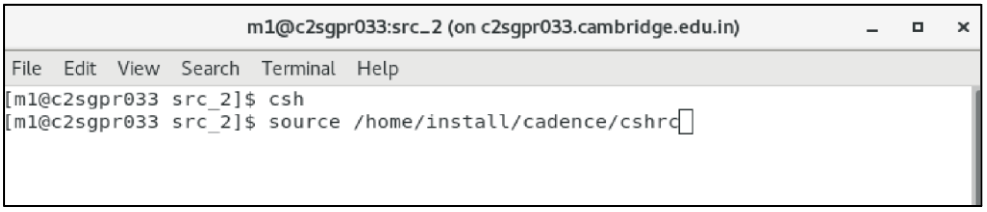


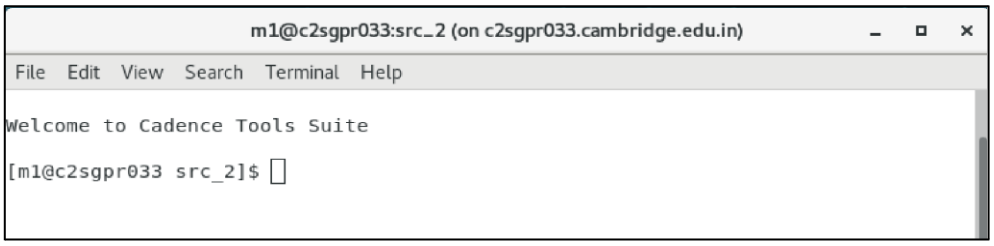
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Follow up these Steps:-

- Open Terminal window
    - Login: - ssh –X (username) @ server ip
    - Password:-XXXXXXXX
- To work in Cadence Tool Suite
    - csh
    - Source /home/install/cadence/cshrc







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RTL SIMULATION FLOW

1. Create your RTL Design(Verilog/System Verilog) and Simulate using the Cadence Xcelium tool

   a. Follow up these Steps:-

2. Open a Work Directory **1_CIT_SAR_SIMULATION**

   I. To work in Cadence Tool Suite
   - csh
   - source /home/install/cadence/cshrc

3. To open and run Xcelium
   i. xrun -v sar.v sar_tb.v –access +rwc -gui

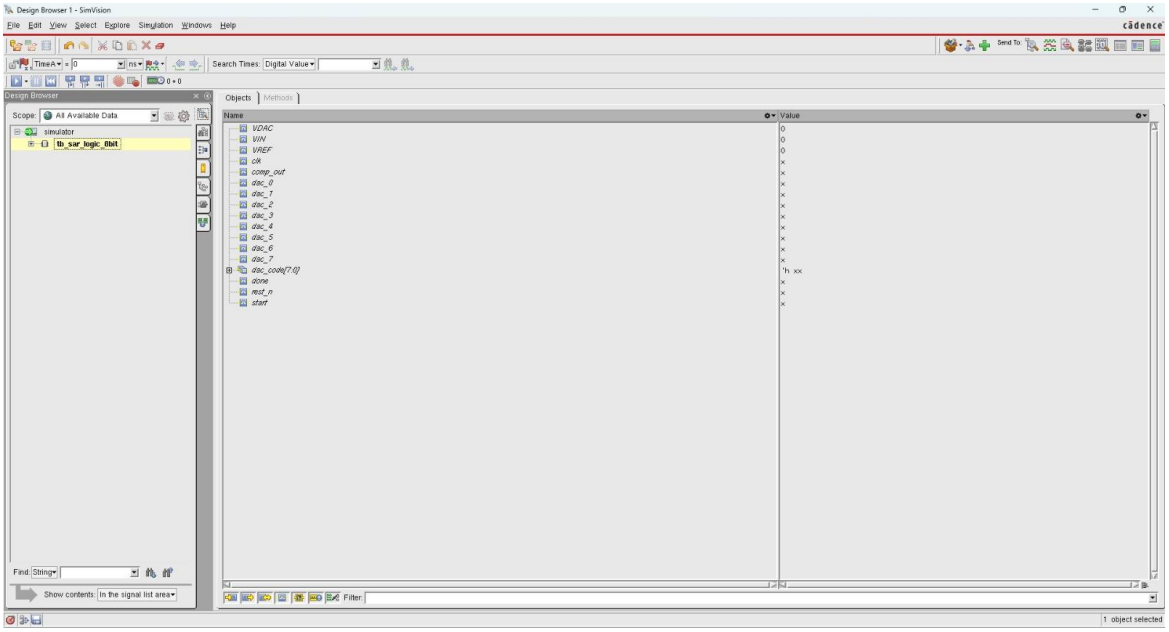
```
[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ csh
[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ source /home/install/cadence/cshrc

Welcome to Cadence Tools Suite

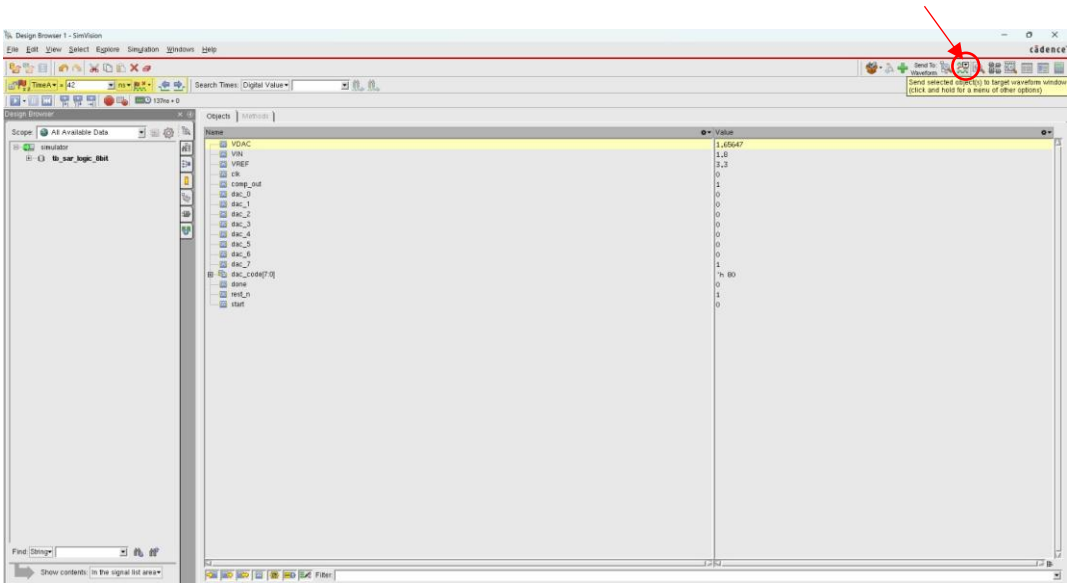
[student@c2sgpr050 1_CIT_SAR_SIMULATION]$ xrun sar.v sar_tb.v -access +rwc -gui
```

4. After running this command the tool will invoke in Simvison where you can see the tb_sar_logic_8bit

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5. Click the tb_sar_logic_8bit and send this to waveform window and you can see a waveform window like this



6. After the waveform window opened click run to see the simulation in this you can check the functionality of the design

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7. After verifying the functionality proceed to synthesis

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SYNTHESIS

1. As you confirm your RTL design is functionally correct, you need to synthesized the design and create gate-level-netlist(.v synthesized file).For synthesis we use Cadence Genus tool.

2. Follow up these Steps:-

a. Open a Work Directory **2_CIT_SAR_SYNTHESIS**

II. To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 2_CIT_SAR_SYNTHESIS]$ csh
[student@c2sgpr050 2_CIT_SAR_SYNTHESIS]$ source /home/install/cadence/cshrc

Welcome to Cadence Tools Suite

[student@c2sgpr050 2_CIT_SAR_SYNTHESIS]$ genus
```

i. script is provided for genus

ii. sar_run.tcl and sar.sdc

```
### Template Script for RTL->Gate-Level Flow
## Preset global variables and attributes
################################################################################
set DESIGN sar_logic_8bit
set GEN_EFF medium
set MAP_OPT_EFF high
set clockname clk
set DATE [clock format [clock seconds] -format "%b%d-%T"]
set _OUTPUTS_PATH output
set _REPORTS_PATH report
set _LOG_PATH /home/soujanya/sar_pwm/2_CIT_SAR_synth/logs
set LOG_PATH logs${DATE}
set_db / .init_lib_search_path {/opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ss/ ./opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ff/}
set_db auto_ungroup none
set_db / .information_level 7
################################################################
## Library setup
################################################################
set_db / .library {/opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ss/tsl18fs120_scl_ss.lib /opt/sclpdkv3/scl180/iopad/cio150/6M1L/liberty/tsl18cio150_max.lib
/opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ff/tsl18fs120_scl_ff.lib /opt/sclpdkv3/scl180/iopad/cio150/6M1L/liberty/tsl18cio150_min.lib}
read_libs -max_libs {/opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ss/tsl18fs120_scl_ss.lib /opt/sclpdkv3/scl180/iopad/cio150/6M1L/liberty/tsl18cio150_max.lib} \
-min_libs {/opt/sclpdkv3/scl180/stdcell/fs120/6M1L/liberty/lib_flow_ff/tsl18fs120_scl_ff.lib /opt/sclpdkv3/scl180/iopad/cio150/6M1L/liberty/tsl18cio150_min.lib}
################################################################
## Load Design
################################################################
puts "load RTL"
#set FILE_LIST sar_adc_pwm_top.v sar_logic_8bit.v sar_pwm.v
set FILE_LIST {sar.v}
read_hdl $FILE_LIST
puts "elobrate design"
elaborate $DESIGN
puts "Runtime & Memory after 'read_hdl'"
time_info Elaboration
check_design -unresolved
################################################################
## Constraints Setup
################################################################
read_sdc sar.sdc
path_adjust -from [all_inputs] -to [all_outputs] -delay -1300 -name PA_I2O
path_adjust -from [all_inputs] -to [all_register] -delay -1500 -name PA_I2C
path_adjust -from [all_register] -to [all_outputs] -delay -1500 -name PA_C2O
```


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3. To open and run genus, open terminal and type genus the tool will invoke in terminal mode after it is opened type source sar_logic_8bit_run.tcl
    i. genus
    ii. sar_run.tcl

```
[student@c2sgpr050 2_CIT_SAR_SYNTHESIS]$ genus
TMPDIR is being set to /tmp/genus_temp_12976_c2sgpr050.cambridge.edu.in_student_QoLBip
Cadence Genus(TM) Synthesis Solution.
Copyright 2021 Cadence Design Systems, Inc. All rights reserved worldwide.
Cadence and the Cadence logo are registered trademarks and Genus is a trademark
of Cadence Design Systems, Inc. in the United States and other countries.

Version: 20.11-s111_1, built Mon Apr 26 02:27:38 PDT 2021
Options:
Date:    Mon Sep 28 17:52:42 2026
Host:    c2sgpr050.cambridge.edu.in (x86_64 w/Linux 3.10.0-1160.el7.x86_64) (4cores*4cpus*1physical cpu*Intel(R) Core(TM) i5-7500 CPU @ 3.40GHz 6144KB) (16188408KB)
PID:     12976
OS:      Red Hat Enterprise Linux Server release 7.9 (Maipo)

Checking out license: Genus_Synthesis

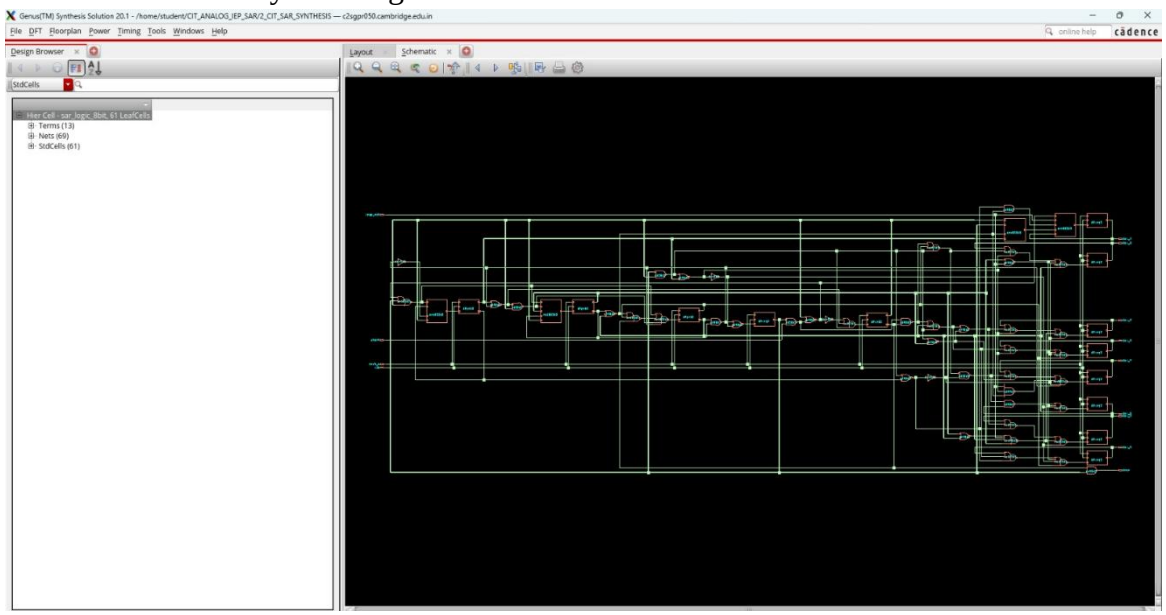

**********************************************************************************************************
**********************************************************************************************************



Loading tool scripts...
Finished loading tool scripts (9 seconds elapsed).

WARNING: This version of the tool is 1981 days old.
@genus:root: 1> source sar_run.tcl
```

4. After the synthesis is completed the gui will open as shown in the below figure in that click the hier_cell then a small window will pop up in that click schematic view then in main option to see the schematic view of your design



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5. Type report_timing in the terminal to see the slack of the design

```
@genus:root: 2> report_timing
============================================================
  Generated by:           Genus(TM) Synthesis Solution 20.11-s111_1
  Generated on:           Sep 28 2026  06:01:50 pm
  Module:                 sar_logic_8bit
  Operating conditions:   tsl18cio150_max (worst_case_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

Path 1: MET (441 ps) Setup Check with Pin sar_reg_reg[5]/CP->D
          Group: I2R
     Startpoint: (R) start
          Clock: (R) vir_clk_i
       Endpoint: (F) sar_reg_reg[5]/D
          Clock: (R) clk

                    Capture       Launch
       Clock Edge:+   10000            0
        Drv Adjust:+      0            0
      Src Latency:+   1000         1250
      Net Latency:+      0 (I)        0 (I)
          Arrival:=  11000         1250

            Setup:-    171
      Uncertainty:-    300
      Path Adjust:+  -1500
     Required Time:=   9029
      Launch Clock:-   1250
       Input Delay:-   6000
         Data Path:-   1338
             Slack:=    441

Exceptions/Constraints:
  path_adjust             0             from_vir_clk_i_to_clk_setup_uncertainty
  path_adjust         -1500             PA_I2C
  input_delay          6000             sar.sdc_line_65

#-------------------------------------------------------------------------------------------------------
#  Timing Point    Flags    Arc    Edge   Cell      Fanout Load Trans Delay Arrival Instance
#                                                    (fF)  (ps)  (ps)   (ps)  Location
#-------------------------------------------------------------------------------------------------------
  start           -        -      R     (arrival)      1  7.0  1000     0     7250    (-,-)
  g1456__5122/ZN  -        A2->ZN F     nd12d1         2 13.0   321   212     7462    (-,-)
  g1446__5107/ZN  -        A1->ZN R     nd02d1         2 13.4   270   201     7663    (-,-)
  g1444__5477/ZN  -        A->ZN  F     oai21d1        3 19.6   292   175     7838    (-,-)
  g1432__6161/Z   -        A1->Z  F     or02d1         2 13.8   173   351     8189    (-,-)
  g1419__8428/ZN  -        B1->ZN R     oai21d1        1  6.7   434   269     8458    (-,-)
  g1413__5107/ZN  -        A->ZN  F     oai21d1        1  5.1   382   129     8588    (-,-)
  sar_reg_reg[5]/D <<<     -      F     dfcrq1         1    -     -     0     8588    (-,-)
#-------------------------------------------------------------------------------------------------------
```

6. Type report_area in the terminal to see the total area of the design

```
@genus:root: 3> report_area
============================================================
  Generated by:           Genus(TM) Synthesis Solution 20.11-s111_1
  Generated on:           Sep 28 2026  06:02:08 pm
  Module:                 sar_logic_8bit
  Operating conditions:   tsl18cio150_max (worst_case_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

   Instance    Module  Cell Count  Cell Area  Net Area   Total Area  Wireload
-----------------------------------------------------------------------------
sar_logic_8bit               61   1636.970    23.399     1660.369  4000 (S)
  (S) = wireload was automatically selected
@genus:root: 4>
```


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7. Next type report_power in the terminal to see the total power of the design

```
@genus:root: 4> report_power
Info    : Joules engine is used. [RPT-16]
        : Joules engine is being used for the command report_power.
Info    : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
        : sar_logic_8bit
Info    : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info    : ACTP-0001 Activity propagation ended for stim#0
Info    : PWRA-0001 [PwrInfo] compute_power effective options
        : -mode : vectorless
        : -skip_propagation : 1
        : -frequency_scaling_factor : 1.0
        : -use_clock_freq : stim
        : -stim :/stim#0
        : -fromGenus : 1
Info    : ACTP-0001 Timing initialization started
Info    : ACTP-0001 Timing initialization ended
Info    : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
        : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
        : flow. Ignoring frequency scaling.
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
        : of power analysis, ignored this option.
Info    : PWRA-0002 Started 'vectorless' power computation.
Info    : PWRA-0002 Finished power computation.
Info    : PWRA-0007 [PwrInfo] Completed successfully.
        : Info=6, Warn=2, Error=0, Fatal=0
Instance: /sar_logic_8bit
Power Unit: W
PDB Frames: /stim#0/frame#0
  ------------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     1.18549e-08  9.59161e-05  5.06734e-05  1.46601e-04  74.70%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     5.69551e-09  1.80385e-05  2.42355e-05  4.22797e-05  21.54%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  7.37456e-06  7.37456e-06   3.76%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ------------------------------------------------------------------------
    Subtotal     1.75504e-08  1.13955e-04  8.22835e-05  1.96256e-04 100.00%
  Percentage           0.01%       58.06%       41.93%      100.00% 100.00%
  ------------------------------------------------------------------------
```

8. Now type exit in the terminal.

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INNOVUS

1. Now as synthesis is done we move for Physical Design using Cadence Innovus

2. **Follow up these Steps:-**

. Open a Work Directory **3_CIT_SAR_FLOORPLAN**

To work in Cadence Tool Suite

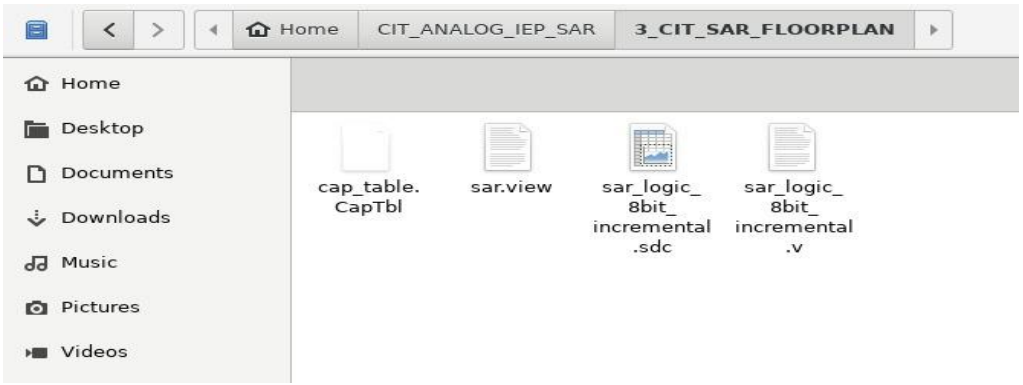
- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 3_CIT_SAR_FLOORPLAN]$ csh
[student@c2sgpr050 3_CIT_SAR_FLOORPLAN]$ source /home/install/cadence/cshrc

Welcome to Cadence Tools Suite

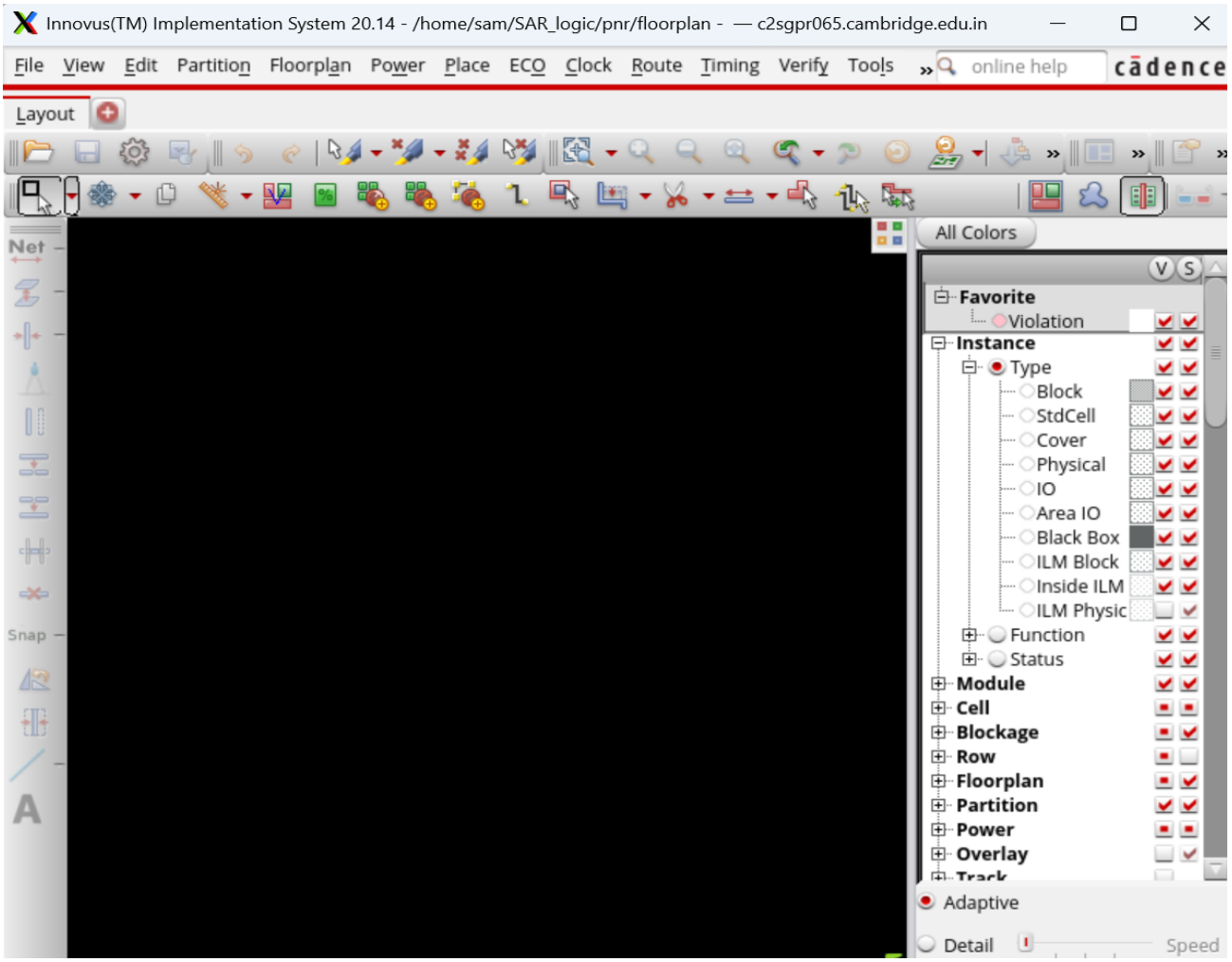
[student@c2sgpr050 3_CIT_SAR_FLOORPLAN]$ innovus
```

1. Files required for PnR from the output files of synthesis
   i. incremental.v(Netlist)
   ii. incremental.sdc
   iii. design.view(MCMM)
   iv. .capTbl file
   v. Design.view (MCMM)



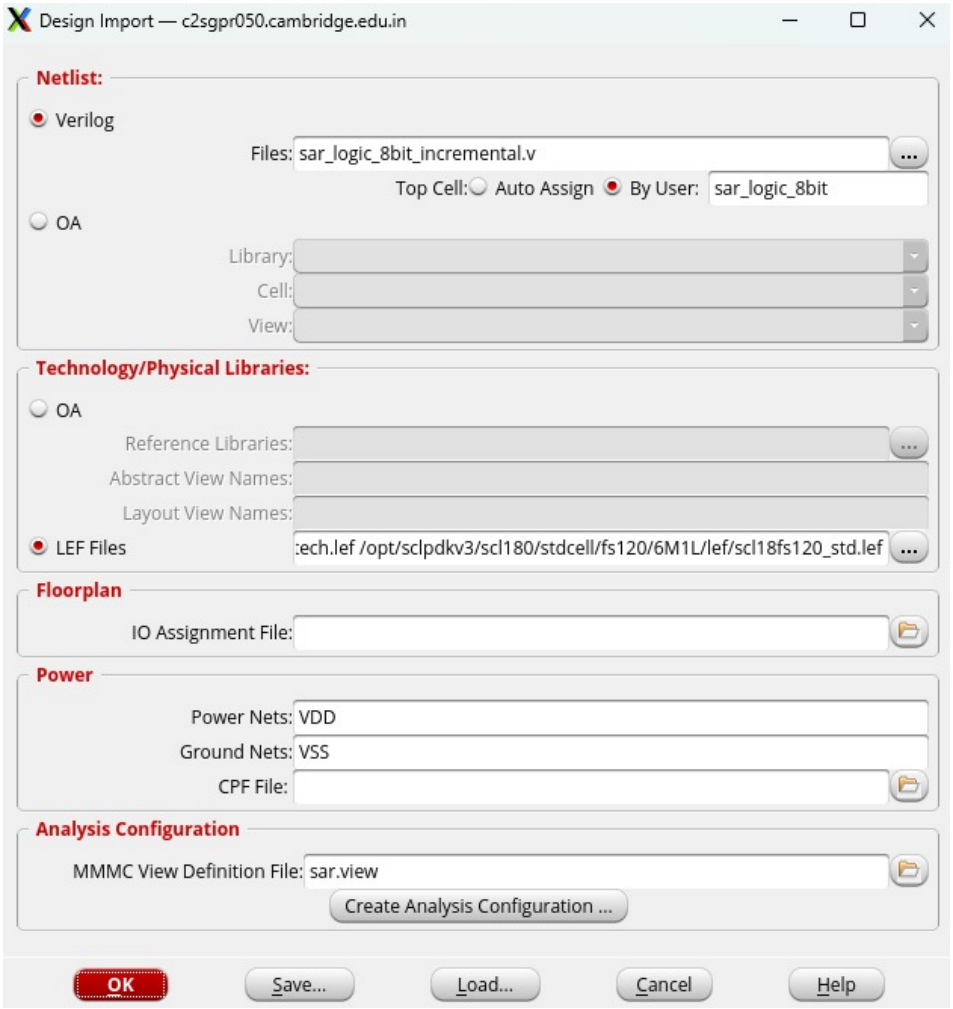
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2. After invoking the Innovus tool you can see the gui as shown in the figure



3. Now click the design import option in the tool to import
    i. netlist.v
    ii. lef files
    iii. mcmm.view file

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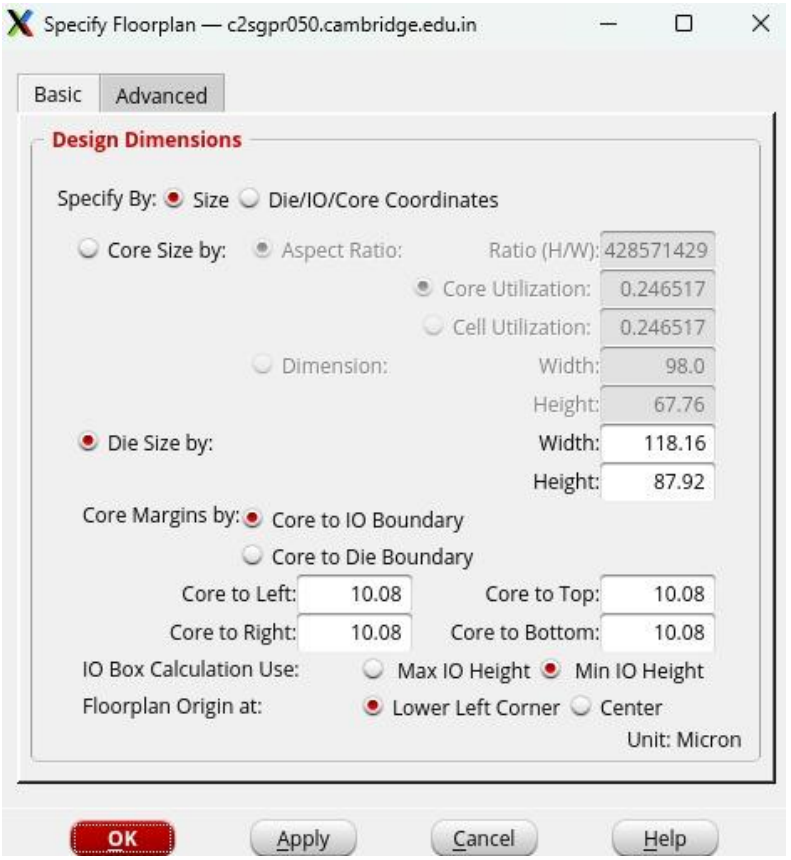


4. After importing all the required files click ok

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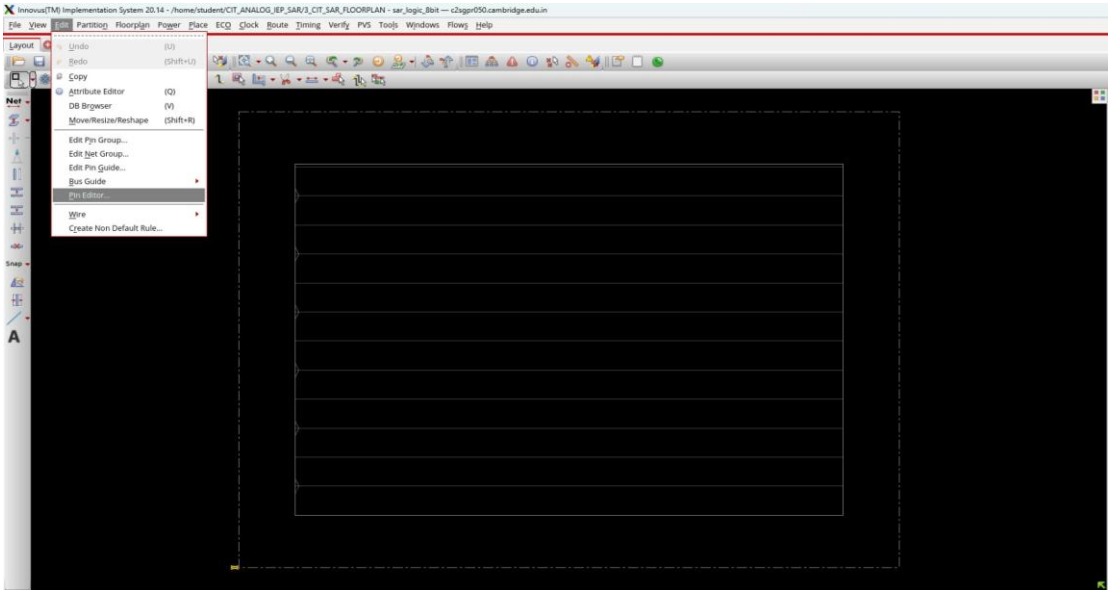
FLOORPLAN

1. Now all the files are imported lets proceed to Floorplan which is a crucial stage in Physical design were we can define the shape, IO pad, utilization rate, aspect ratio, boundary and macro information
2. Click the Floorplan in gui then specify the floorplan, do the same as in the given figure
    i. Select die size and give width & height as 118 & 88
    ii. Set core to io boundary as 10 for all sides

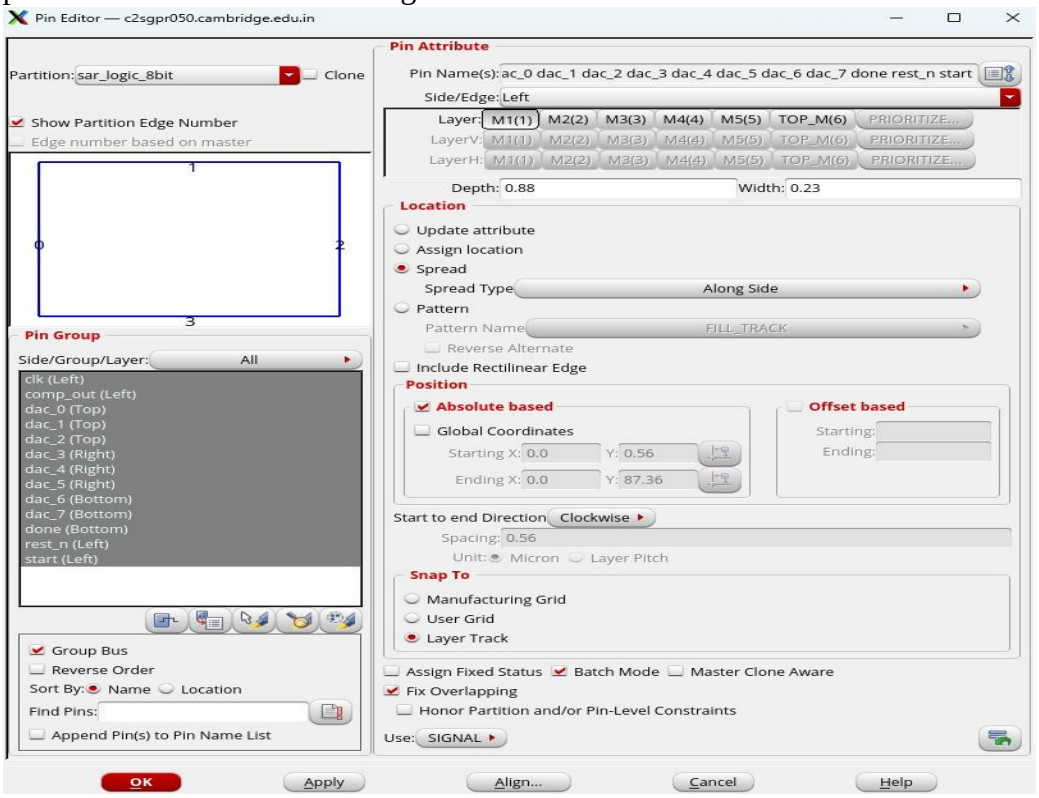


3. Click ok then you can see the floorplan of the design as shown in the above figure
4. Now Click Edit there you can see the pin editor as option click ok, you can see below figure add pins for all sides

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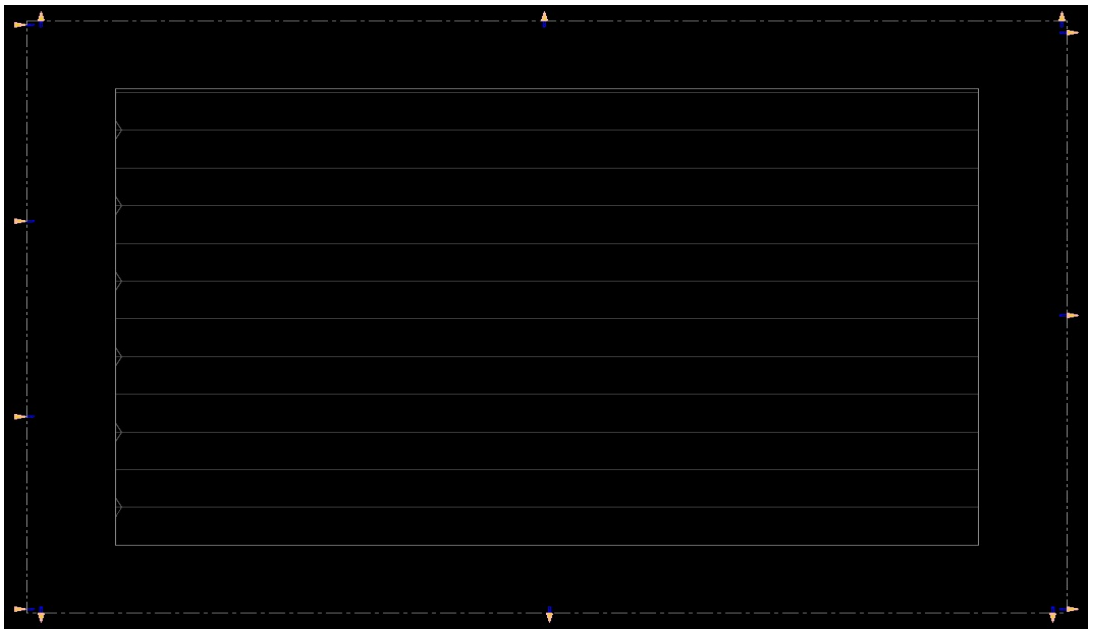


5. Now we need to add pins for all sides, as shown in figure below, for that go to pin group select pins for all sides as shown in figure

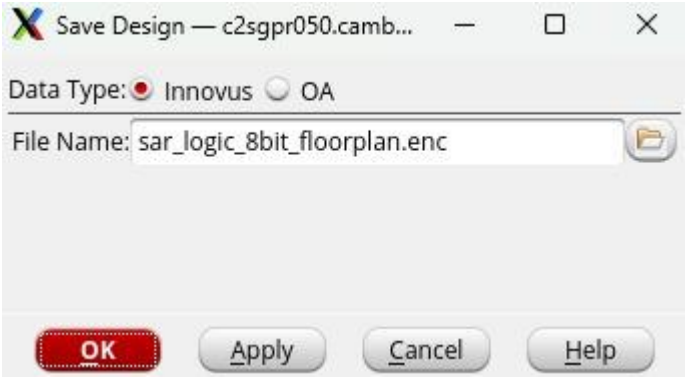


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6. Now we can see the floorplan design



7. Now save the design using the option in the file you can see save design option click that and a window will pop up like this set data type as Innovus option and save the design as floorplan.enc



8. After saving open terminal and type exit to exit the innovus
9. For each stage you need to save the file and exit the tool and need to invoke tool at every stage

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POWERPLAN

Follow up these Steps:-

Open a Work Directory **4_CIT_SAR_POWERPLAN**

To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

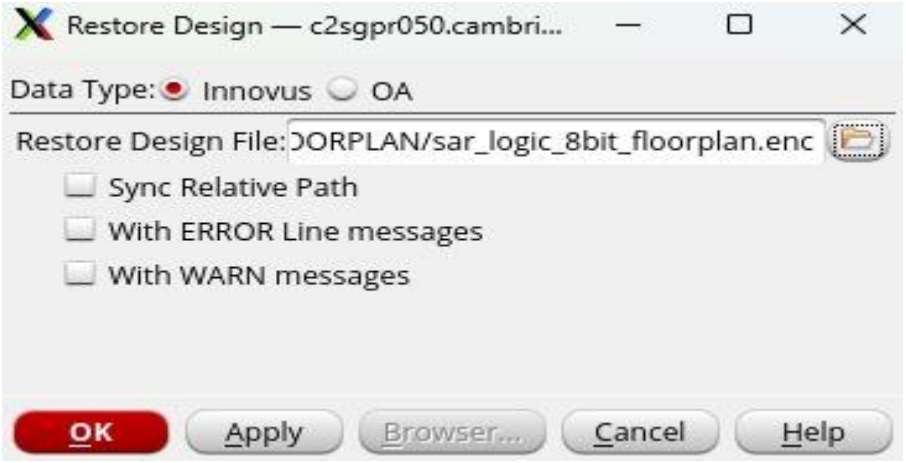
```
[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ csh
[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ source /home/install/cadence/cshrc

Welcome to Cadence Tools Suite

[student@c2sgpr050 4_CIT_SAR_POWERPLAN]$ innovus
```

1. Power plan is part of floorplan were we will define the power rings, stripes, rails and vias which will give the power to our design
2. Open terminal and type innovus to invoke the tool

3. Now we need to restore the design which we saved as sar_logic_8bit_floorplan.enc, for that click file option in gui select the restore design option and next follow as the below figure



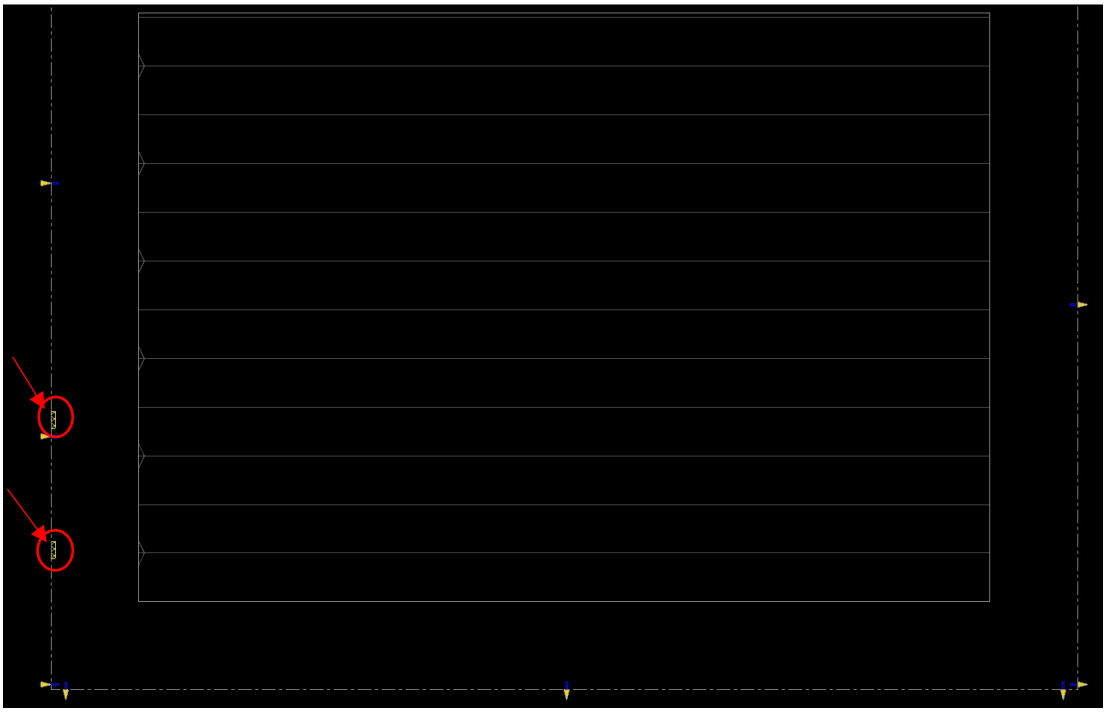
4. Now we need connect the globalNets and create pin for VDD and VSS, so type the given commands in the below figure after this only you should add rings, stripes and standard cell rows.

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```
globalNetConnect VDD -type pgpin -pin VDD -all
globalNetConnect VSS -type pgpin -pin VSS -all
globalNetConnect VDD -type tiehi
globalNetConnect VSS -type tielo


# VDD Pin (Y = 15.0 to 17.0)
createPGPin VDD -net VDD -geom M4 0.30 15.0 0.5 17.0

# VSS Pin (Y = 30.0 to 32.0)
createPGPin VSS -net VSS -geom M4 0.30 30.0 0.5 32.0
```



5. After opening the design let's do power plan, we will start from adding rings first, do the same as in the below figure

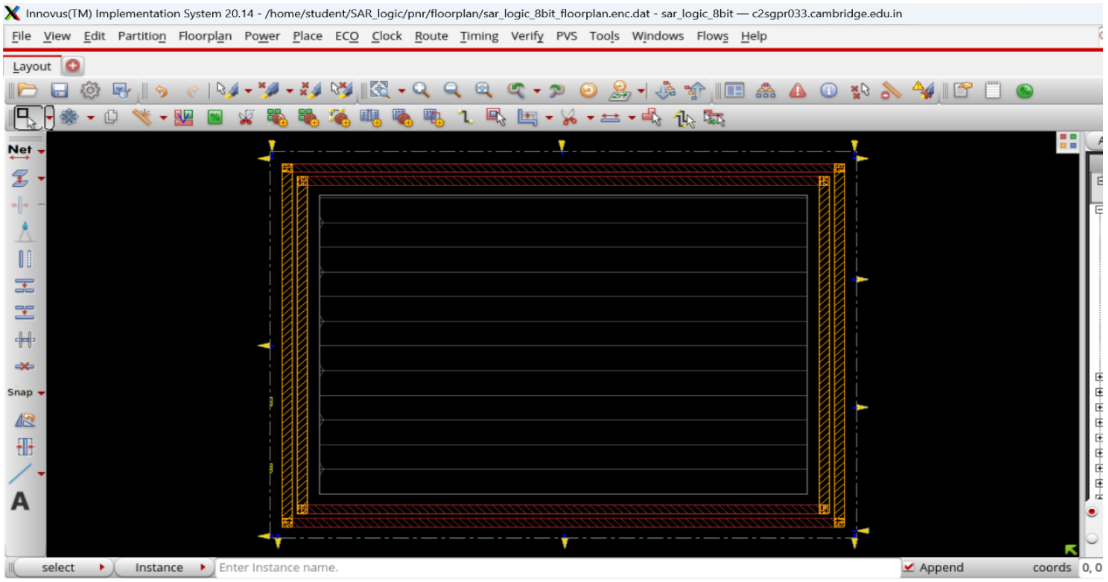
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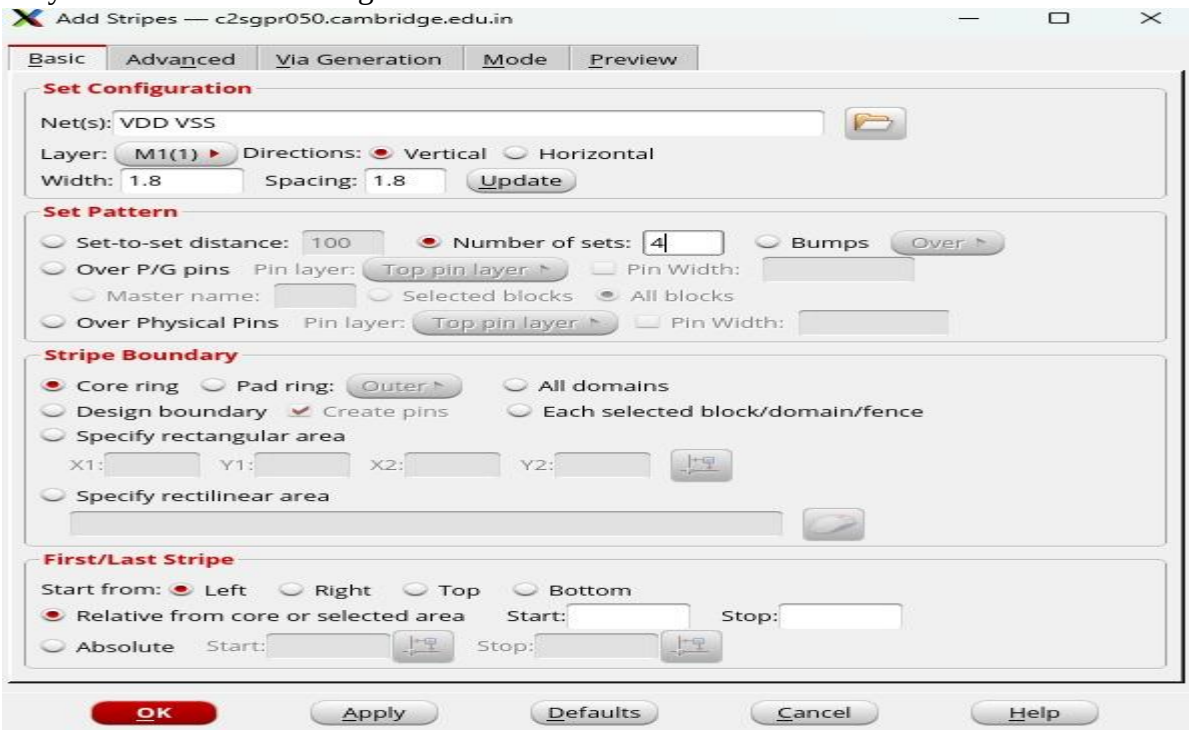
6. Click the Net(s) option and add VDD & VSS as shown in the above figure

7. Now we will set the layers, width and spacing for the rings and set offset (channel in the center) option ok and click ok as shown in the below figure

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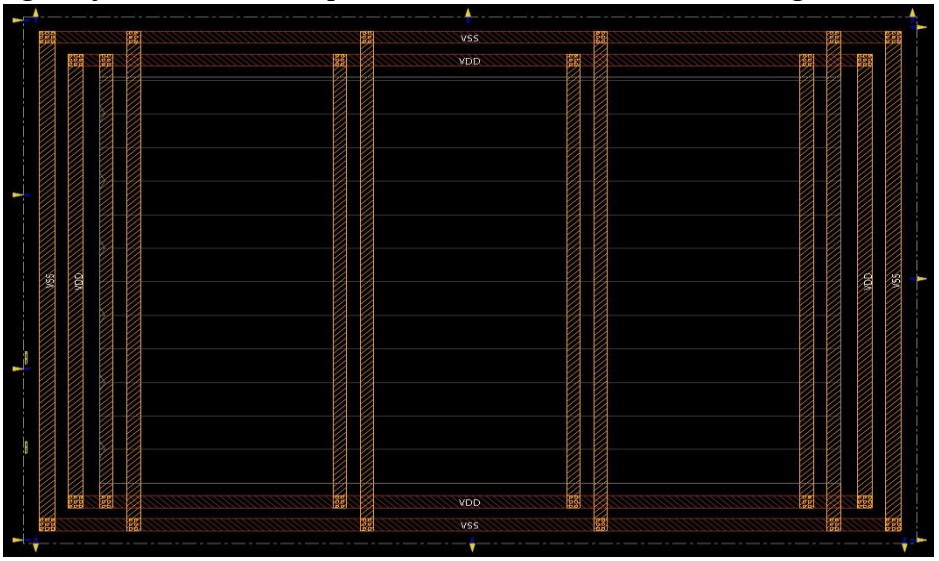


8. Now we need to create a stripes, for that in the gui click Power option then select the add stipes option you can see the below figure in the Innovus window

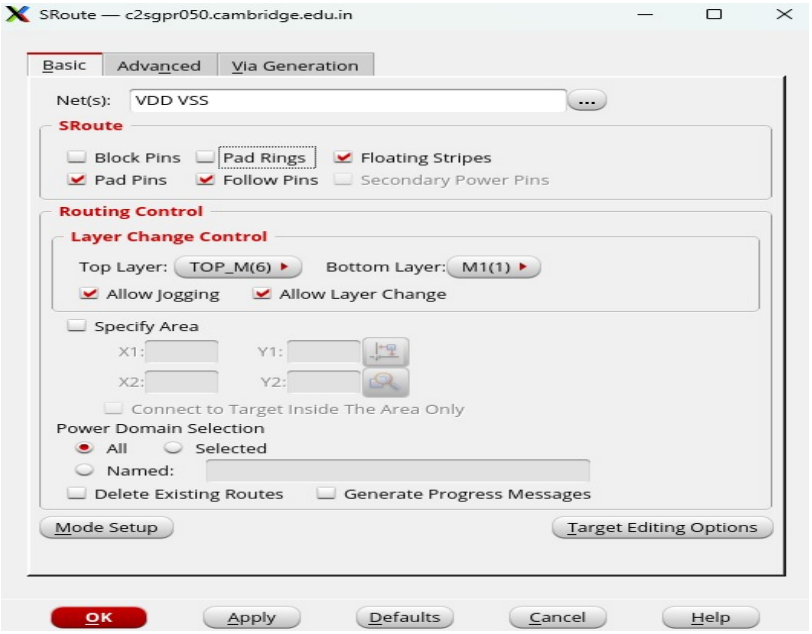


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9. Now do the same for adding nets VDD and VSS and set the metal layers, width, space as shown in the above figure
10. After clicking ok you can see the stipes created as shown in the below figure

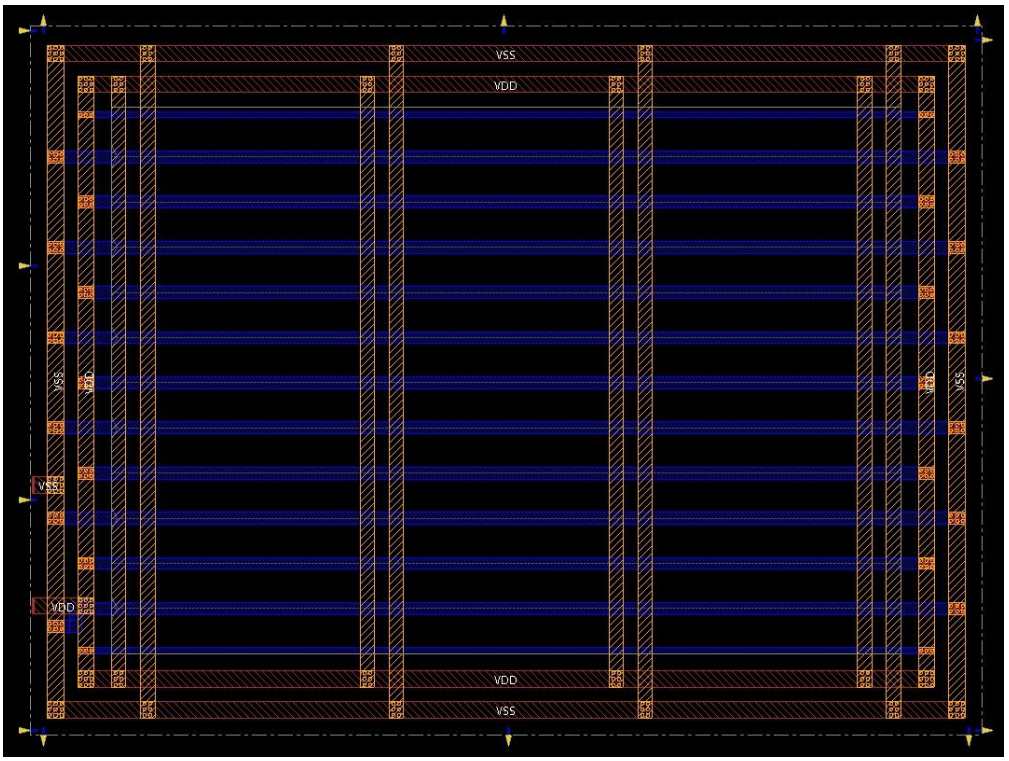


11. Now we need to do Sroute for the standard cell rows for that click Route option in the gui in that select special route option a window will pop up set the same settings shown in the below figure and then click ok

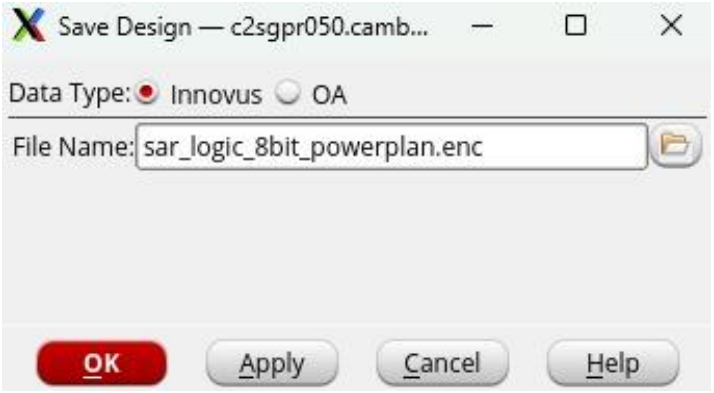


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12. After click ok the sroute will start then you can see the sroute in the below figure



13. Now save the design as powerplan.enc as shown in the below figure by clicking ok



14. After saving file open terminal and type exit

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PLACEMENT

Follow up these Steps:-

. Open a Work Directory **5_CIT_SAR_POWERPLAN**

To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ csh
[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ source /home/install/cadence/cshrc

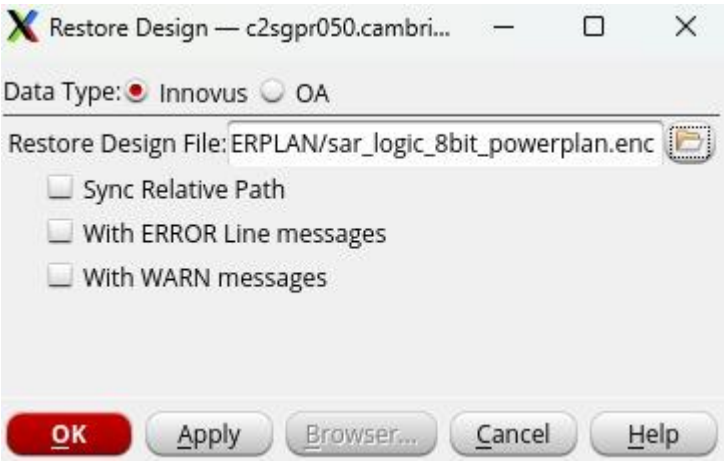
Welcome to Cadence Tools Suite

[student@c2sgpr050 5_CIT_SAR_PLACEMENT]$ innovus
```

1. Placement is where all the standard cells(gates,flip_flops), macros are placed inside the core

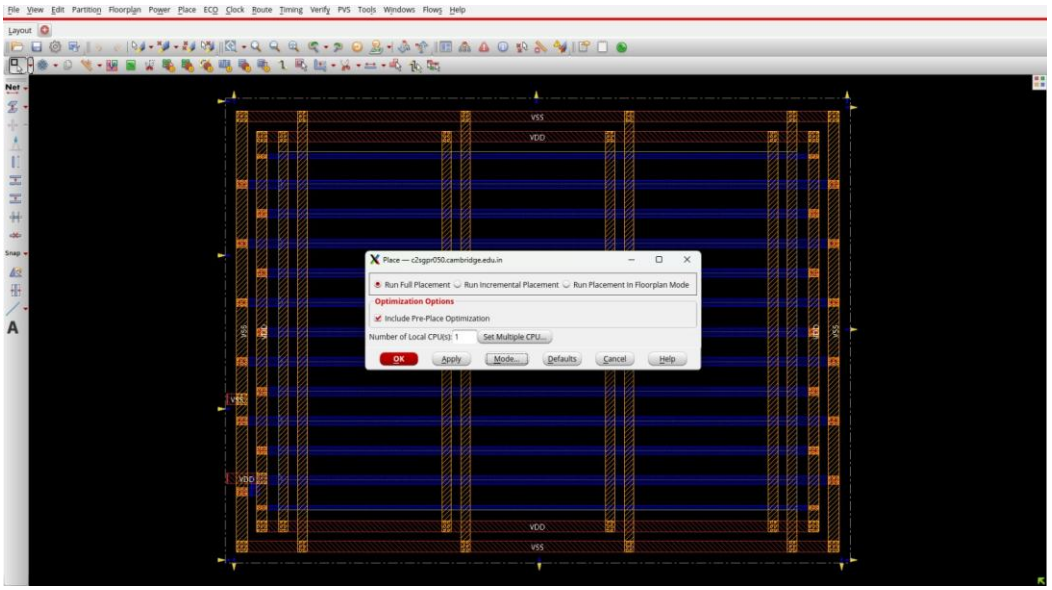
2. Open terminal and type innovus to invoke tool

3. Now restore the saved design from powerplan stage as shown in the below figure

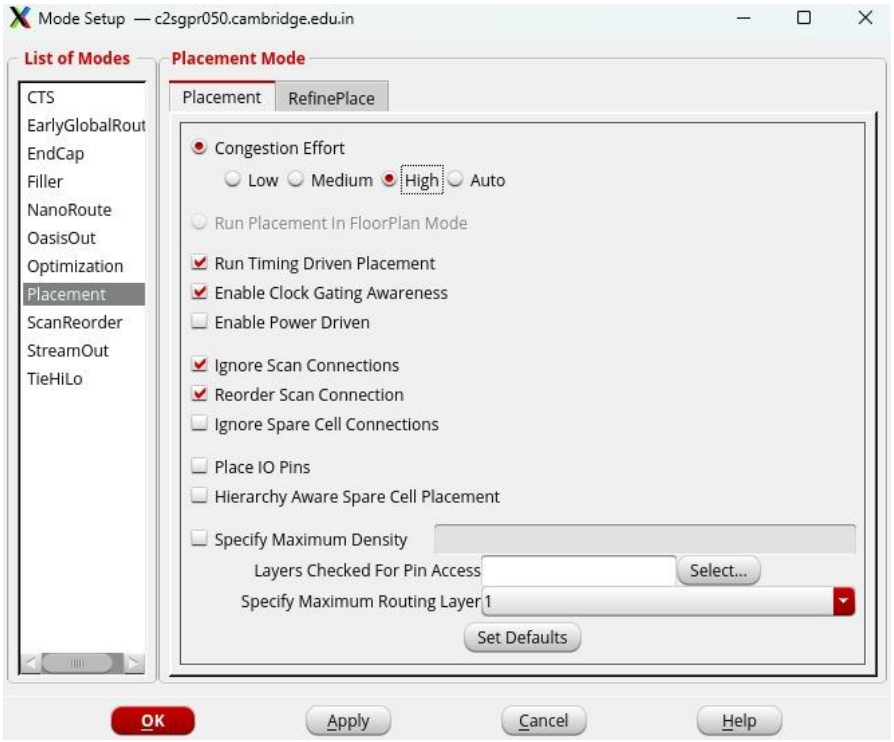


4. Now click the Place option in the gui for doing placement in that click the placement standard cell option a window will pop as shown in the below figure and select all the options same as the below figure

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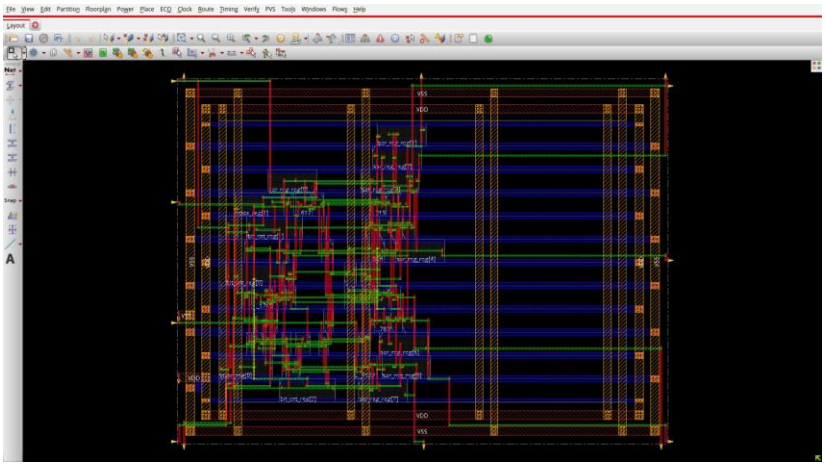
5 select the mode and select the same options as shown in the below figure and click ok



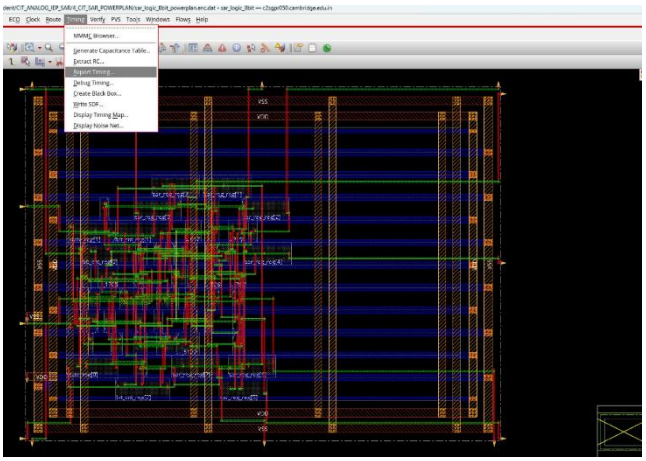
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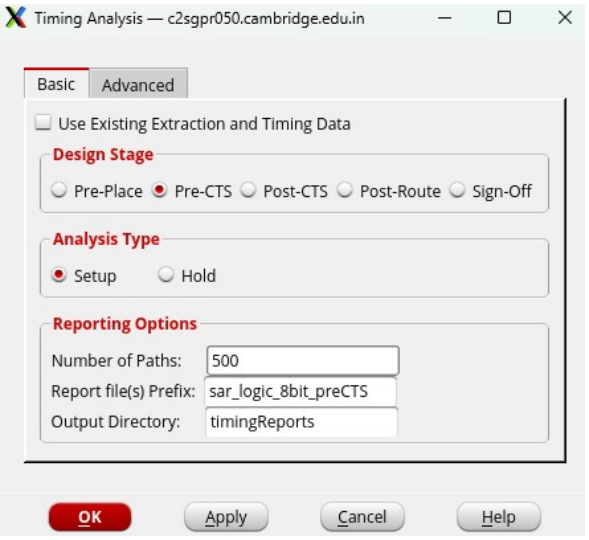
6 Now click ok then placement will happen

7 After the placement of standard cells is done you can see the placement as in the below figure



6. click Timing option in the gui and select the report timing option you can see the window pop up as shown in the below figure and do the same as shown to check if any violations like setup, max transitions, capacitance in the design.





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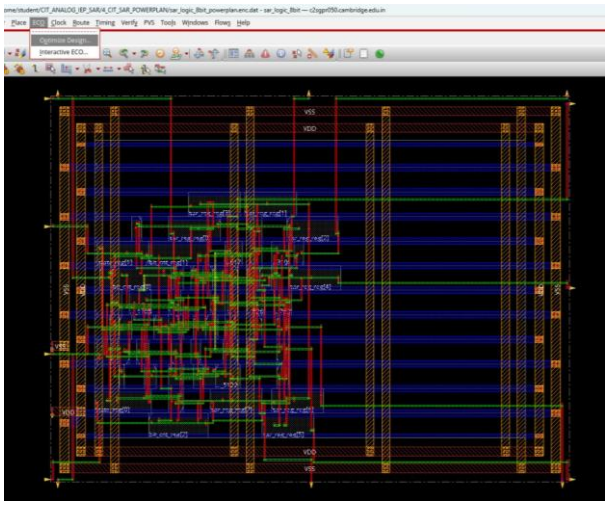
```
------------------------------------------------------------------
         timeDesign Summary
------------------------------------------------------------------

Setup views included:
 worst

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  2.111  |  6.580  |  2.111  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   22    |   13    |   21    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+
```

7. Now if there are any violations fix the violations before proceeding to CTS, for that click ECO option the in the gui and click optimization a window will pop up and do the same as shown in the below figure





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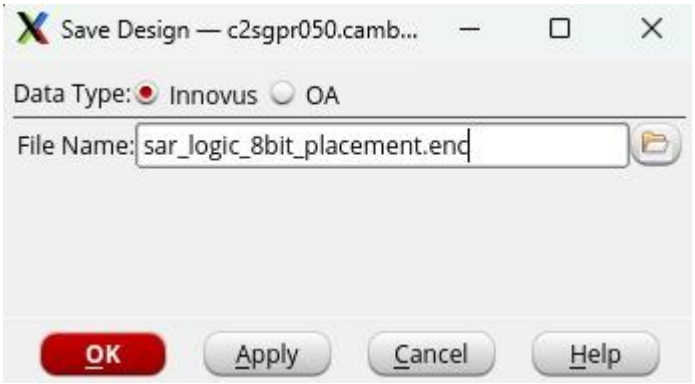
8. you can see there is no violations so let's save the design as sar_logic_8bit_placement.enc in the save design Innovus option in gui

```
------------------------------------------------------------------
     optDesign Final Summary
------------------------------------------------------------------

Setup views included:
 worst

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  2.111  |  6.580  |  2.111  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   22    |   13    |   21    |
+--------------------+---------+---------+---------+

+----------------+-----------------------------------+------------------+
|                |              Real                 |       Total      |
|    DRVs        +------------------+----------------+------------------|
|                |  Nr nets(terms)  |  Worst Vio     |  Nr nets(terms)  |
+----------------+------------------+----------------+------------------+
|   max_cap      |      0 (0)       |   0.000        |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000        |      0 (0)       |
|   max_fanout   |      0 (0)       |     0          |      0 (0)       |
|   max_length   |      0 (0)       |     0          |      0 (0)       |
+----------------+------------------+----------------+------------------+
```



9. After saving open terminal and type exit to exit innovus

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CLOCK TREE SYNTHESIS

Follow up these Steps:-

Open a Work Directory **6_CIT_SAR_CTS**

To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 6_CIT_SAR_CTS]$ csh
[student@c2sgpr050 6_CIT_SAR_CTS]$ source /home/install/cadence/cshrc

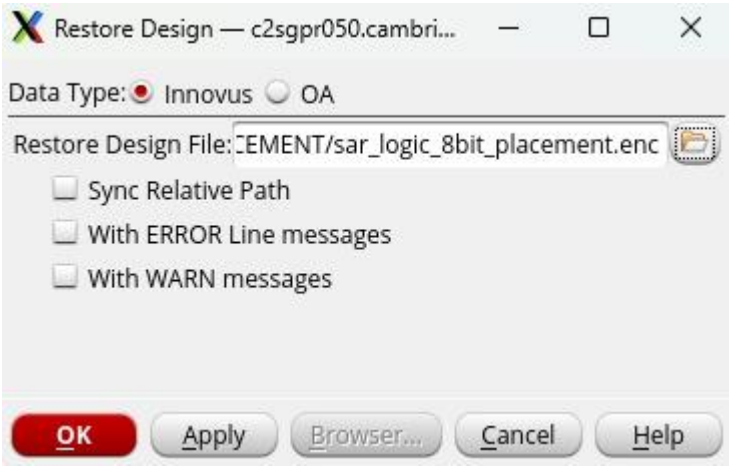
Welcome to Cadence Tools Suite

[student@c2sgpr050 6_CIT_SAR_CTS]$ innovus
```

1. CTS is where the clock signal is distributed from the clock source to all sequential elements with controlled skew and delay.

2. Open terminal and type innovus to invoke tool

3. Now restore the saved design from placement stage as shown in the below figure



4. For doing Clock Tree Synthesis there will be two files included in the directory as shown in the below figure

   i. cts.tcl
   ii. config.tcl

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5. Now we need to source cts.tcl for doing cts the config.tcl is sourced inside cts.tcl

```
##############################################################################
# CTS SCRIPT
#
# Design       : 8-bit SAR ADC + PWM
# Top Module   : sar_adc_pwm_top
# Technology   : SCL 180nm
# Clock        : clk
# Clock Period : 10 ns
# Clock Freq   : 100 MHz
##############################################################################



set init_top_cell sar_adc_pwm_top

set CTS_DIR /home/student/CIT_ANALOG_IEP_SAR/6_CIT_SAR_CTS/



set_global report_timing_format \
    {instance arc net cell slew delay arrival required}



source /home/student/CIT_ANALOG_IEP_SAR/6_CIT_SAR_CTS/config.tcl



puts "================================================="
puts "Creating CCOpt Clock Tree Specification"
puts "Design : $init_top_cell"
puts "Clock  : clk"
puts "Period : 10 ns"
puts "================================================="

create_ccopt_clock_tree_spec \
    -file ${CTS_DIR}/${init_top_cell}_ccopt.spec



source ${CTS_DIR}/${init_top_cell}_ccopt.spec
```

6. Now source the cts.tcl by using command source cts.tcl in the terminal as shown in the below figure

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```
Welcome to Cadence Tools Suite

[student@c2sgpr050 6_CIT_SAR_CTS]$ innovus

Cadence Innovus(TM) Implementation System.
Copyright 2020 Cadence Design Systems, Inc. All rights reserved worldwide.

Version:        v20.14-s095_1, built Mon Apr 19 14:41:42 PDT 2021
Options:
Date:           Mon Sep 28 19:16:13 2026
Host:           c2sgpr050.cambridge.edu.in (x86_64 w/Linux 3.10.0-1160.el7.x86_64) (4cores*4cpus*Intel(R) Core(TM) i5-7500 CPU @ 3.40GHz 6144KB)
OS:             Red Hat Enterprise Linux Server release 7.9 (Maipo)

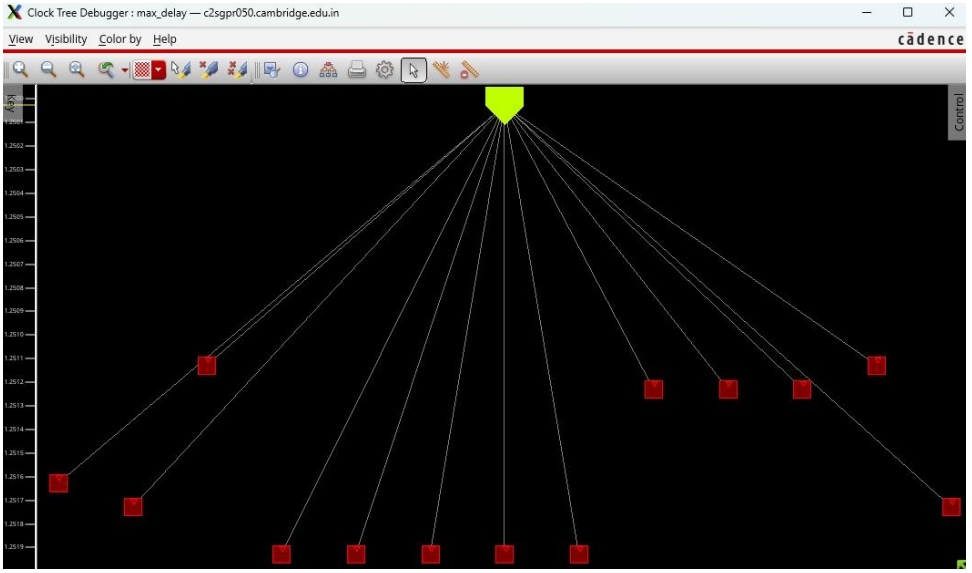
License:
                invs    Innovus Implementation System   20.1    checkout succeeded
                8 CPU jobs allowed with the current license(s). Use setMultiCpuUsage to set your required CPU count.
Create and set the environment variable TMPDIR to /tmp/innovus_temp_14506_c2sgpr050.cambridge.edu.in_student_mGB2eK.

Change the soft stacksize limit to 0.2%RAM (31 mbytes). Set global soft_stack_size_limit to change the value.

**INFO:  MMMC transition support version v31-84

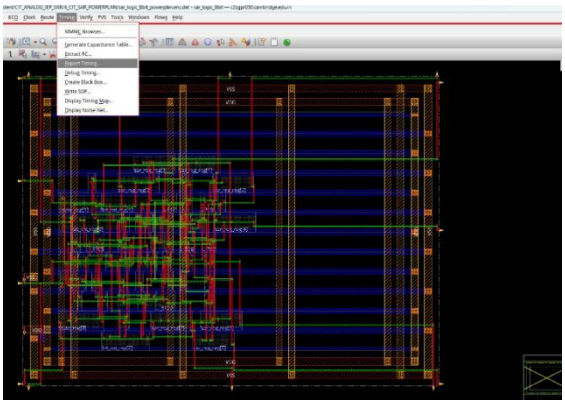
[INFO] Loading PVS 20.11 fill procedures
innovus 1> source cts.tcl
```

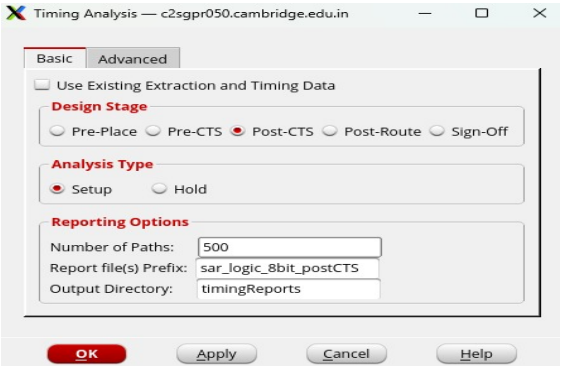
7. After the cts is completed you can see the clock tree debugger as shown in the below figure



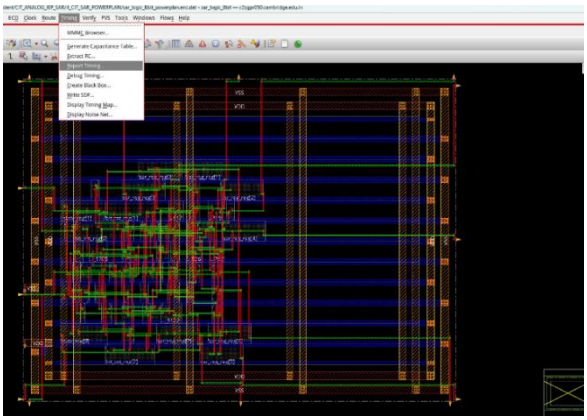
8. Now click Timing option in the gui and select the report timing option you can see the window pop up as shown in the below figure and do the same as shown to check if any violations of setup, hold, max transitions, capacitance in the design.

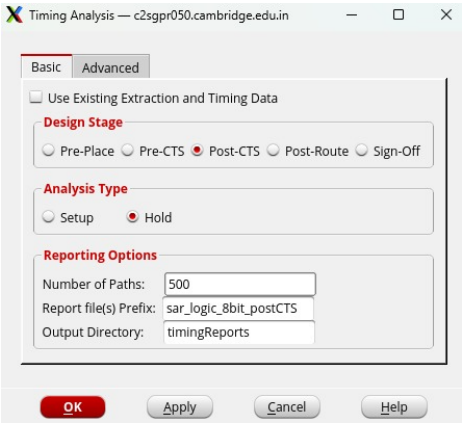
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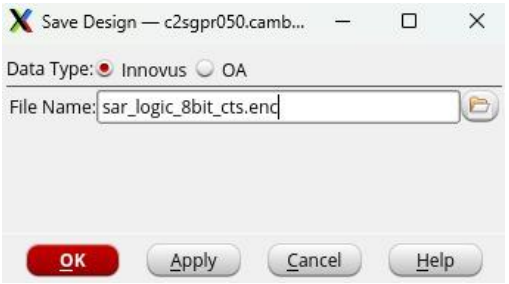


9.Now if see there is hold violation, so we need to fix for that click ECO option the in the gui and click optimization a window will pop up and do the same as shown in the below figure





10. Now you can see there is no violations so let's save the design as cts.enc in the save design Innovus option in gui



11. After saving open terminal and type exit to exit innovus

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ROUTING

Follow up these Steps:-

Open a Work Directory **7_CIT_SAR_ROUTING**

To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 7_CIT_SAR_ROUTING]$ csh
[student@c2sgpr050 7_CIT_SAR_ROUTING]$ source /home/install/cadence/cshrc

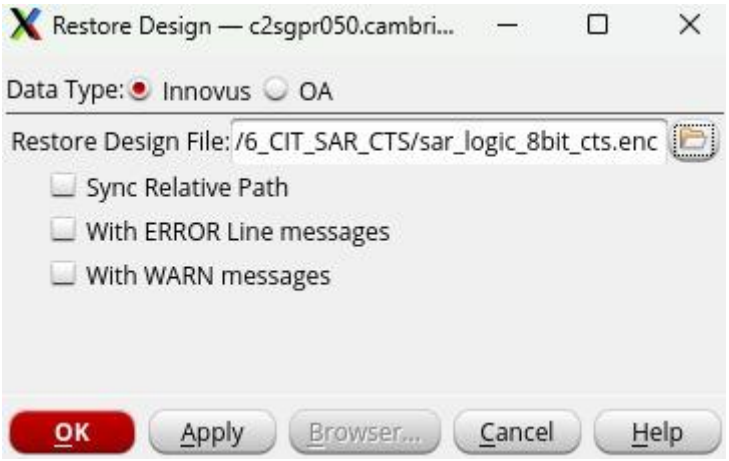
Welcome to Cadence Tools Suite

[student@c2sgpr050 7_CIT_SAR_ROUTING]$ innovus
```

1. Routing is where all placed cells are connected using metal layers according to netlist, the timing, drc should be clean in this stage

2. Open terminal and type innovus to invoke tool

3. Now restore the saved design from cts stage which will be sar_logic_8bit_cts.enc

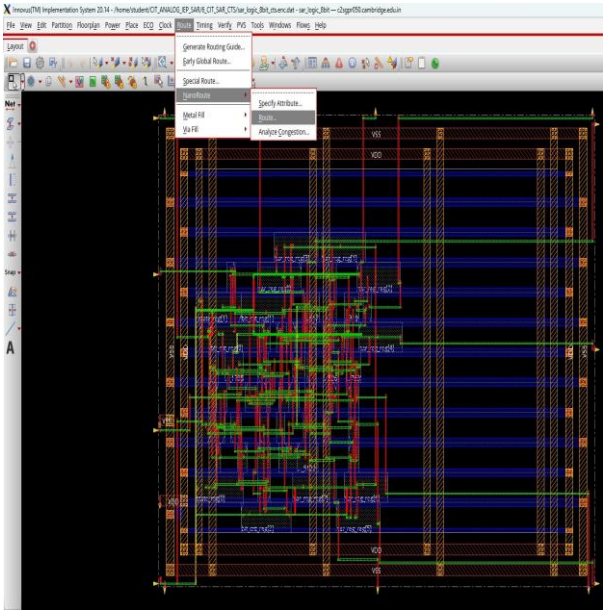


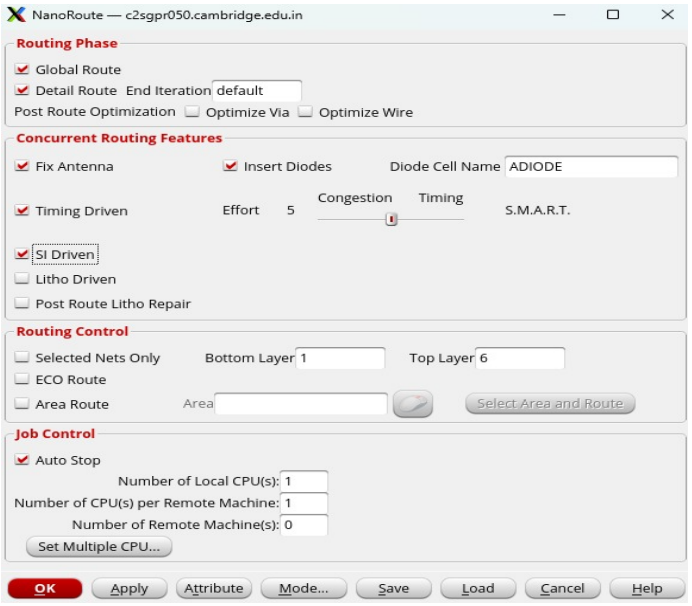
4. Before doing routing run this command in teminal

```
innovus 1> setAnalysisMode -analysisType onChipVariation
```

5. For routing click Route option in the gui where you can see nano route a window will pop up like this, in that select everything same as this figure and type where it's necessary

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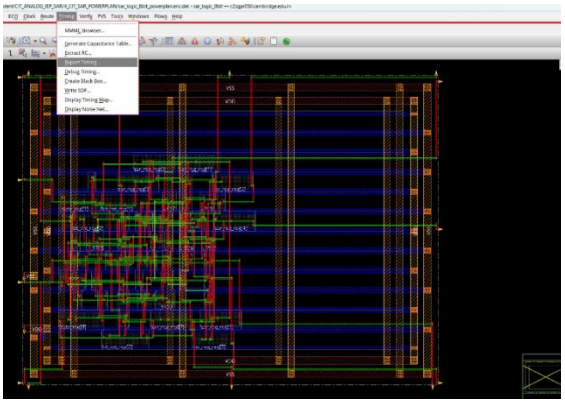


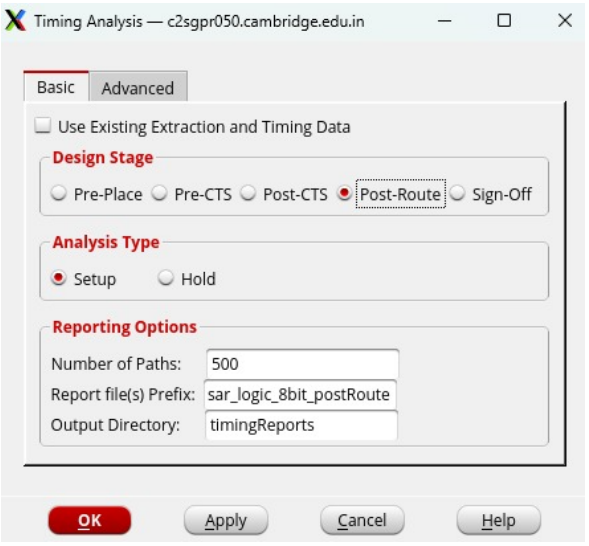


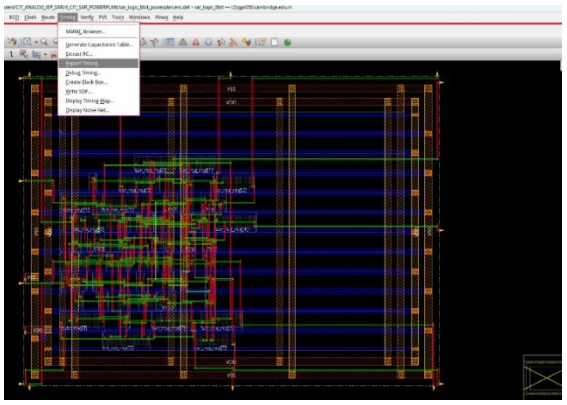
6. We insert ADIODE to avoid antenna affect and set Detail Route End iteration as default

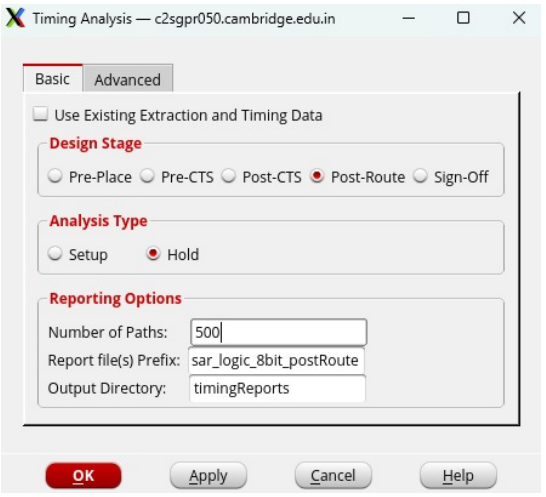
7. Now click Timing option in the gui and select the report timing option you can see the window pop up as shown in the below figure and do the same as shown to check if any violations of setup, hold, max transitions, capacitance in the design.

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8. Now timing is ok hence no need for eco optimization, so let's check if there is any DRC violations in the design for that type verify_drc in the terminal

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```
------------------------------------------------------------------
     optDesign Final SI Timing Summary
------------------------------------------------------------------

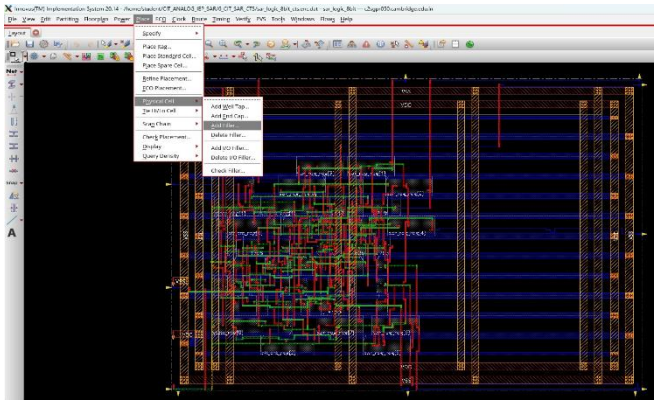
Setup views included:
 worst
Hold views included:
 best

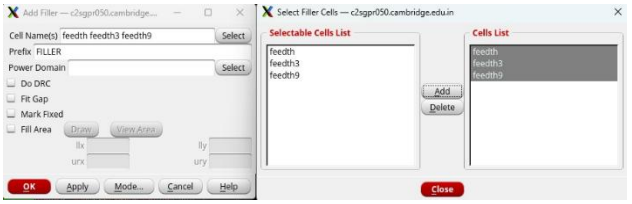
+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  1.476  |  6.787  |  1.476  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|         All Paths:|   22    |   13    |   21    |
+--------------------+---------+---------+---------+

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.042  |  0.042  |  0.364  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|         All Paths:|   22    |   13    |   21    |
+--------------------+---------+---------+---------+

+----------------+-----------------------------------+------------------+
|                |              Real                 |       Total      |
|    DRVs        +------------------+----------------+------------------|
|                |  Nr nets(terms)  |  Worst Vio     |  Nr nets(terms)  |
+----------------+------------------+----------------+------------------+
|   max_cap      |      0 (0)       |    0.000       |      0 (0)       |
|   max_tran     |      0 (0)       |    0.000       |      0 (0)       |
|   max_fanout   |      0 (0)       |      0         |      0 (0)       |
|   max_length   |      0 (0)       |      0         |      0 (0)       |
+----------------+------------------+----------------+------------------+
```

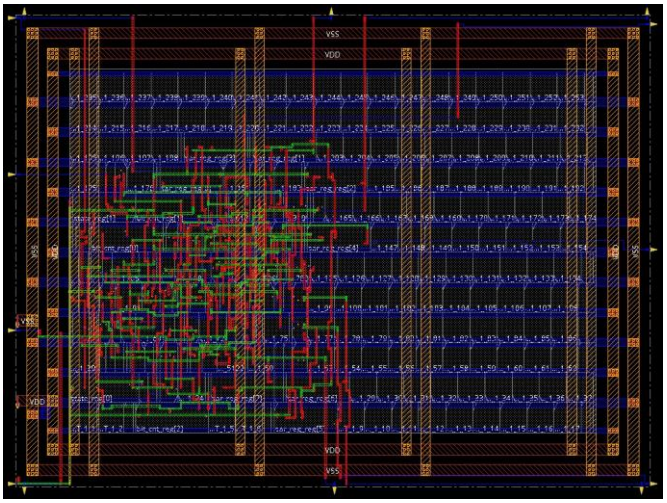
9. Now we need to add filler cells which are used to fill the gaps in the design for continuity for that click Place option in the gui where you can see add filler option after clicking that a window will pop up as shown in the below figure



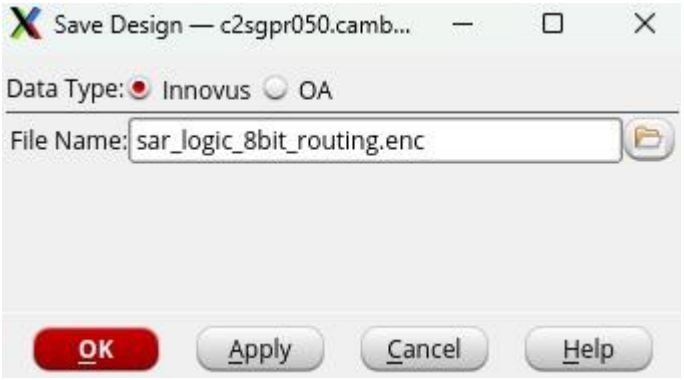


10. Now we can see in the terminal that the fillers are added

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11. Now we can save the design as sar_logic_8bit_routing.enc



12. After saving open terminal and type exit to exit innovus

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SIGNOFF

1. Signoff is the final verification stage where the chip is thoroughly checked to ensure it meets all timing, physical, and manufacturing requirements before tape-out.

2. Open terminal and type innovus to invoke tool

Follow up these Steps:-

Open a Work Directory **8_CIT_SAR_SIGNOFF**

To work in Cadence Tool Suite

- csh
- source /home/install/cadence/cshrc

```
[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ csh
[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ source /home/install/cadence/cshrc

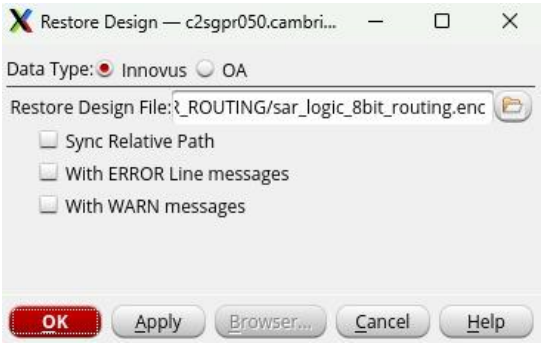
Welcome to Cadence Tools Suite

[student@c2sgpr050 8_CIT_SAR_SIGNOFF]$ innovus
```

3. Signoff is the final verification stage where the chip is thoroughly checked to ensure it meets all timing, physical, and manufacturing requirements before tape-out.

4. Open terminal and type innovus to invoke tool

5. Now restore the saved design from routing stage with filler which will be sar_logic_8bit_routing.enc



6. Steps to be followed in Signoff

   i. Extract Rc
   ii. Verify drc

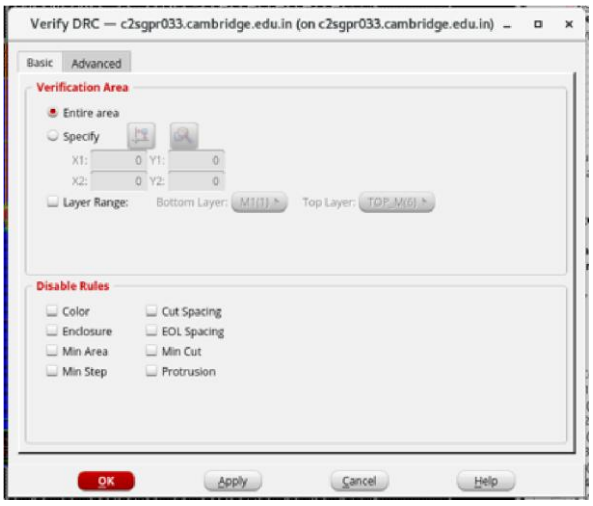
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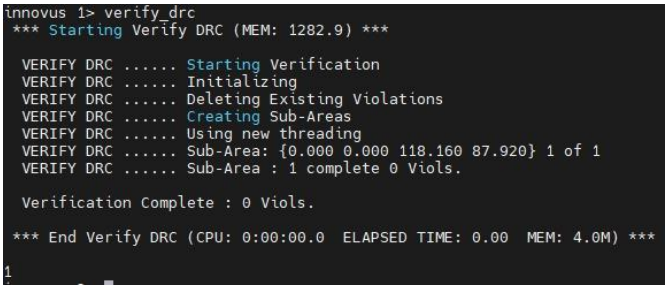
iii. GDS
iv. Save Netlist with PG & without PG
v. Write sdc

7. To extract rc click Timing option in the gui then select extract rc option and select the same options in the window that will pop up as shown in the below figure



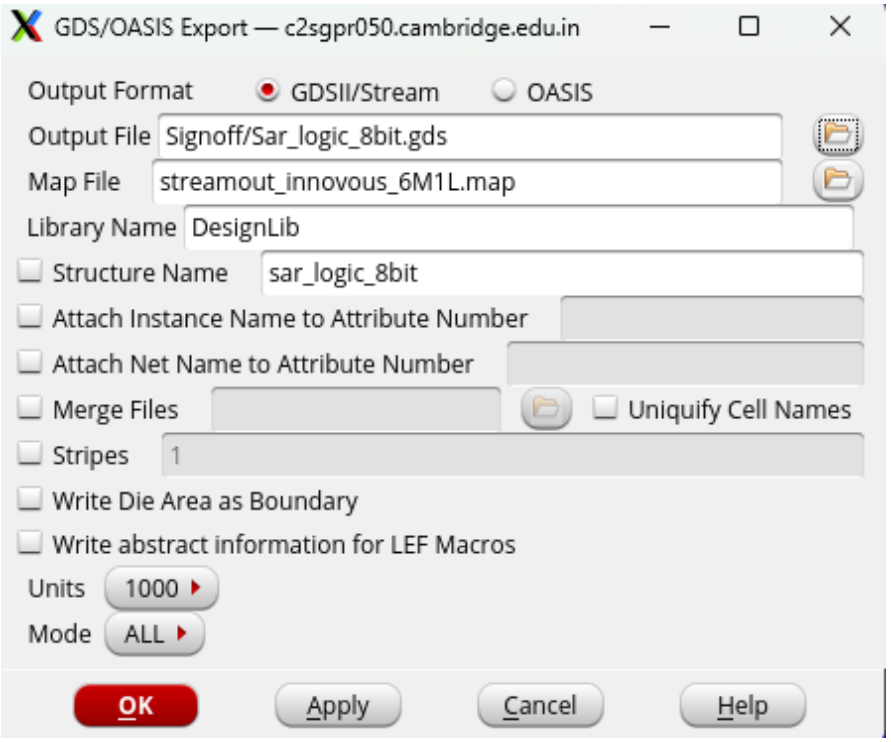
8. To verify drc click the verify option in the gui then select verify drc a window will pop up give the options as same as in the below figure





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9. To save/export the gds, click the file option in the gui select save as in that select gds/oasis then do the same as in the below figure and one important thing to do is correctly map the stream map file and click ok then see the terminal as shown in the below figure




```
Stream Out Information Processed for GDS version 3:
Units: 1000 DBU

Object                             Count
----------------------------------------
Instances                            316

Ports/Pins                            15
    metal layer M1                    13
    metal layer M4                     2

Nets                                 811
    metal layer M1                   145
    metal layer M2                   487
    metal layer M3                   176
    metal layer M4                     3

    Via Instances                    421

Special Nets                          59
    metal layer M1                    41
    metal layer M5                     6
    metal layer TOP_M                 12

    Via Instances                    158

Metal Fills                           0

    Via Instances                     0

Metal FillOPCs                        0

    Via Instances                     0

Metal FillDRCs                        0

    Via Instances                     0

Text                                 15
    metal layer M1                   13
    metal layer M4                    2

Blockages                             0

Custom Text                           0

Custom Box                            0

Trim Metal                            0

######Streamout is finished!
```

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10. Now save netlist, to save netlist with PG and without PG in the terminal type command as shown in the below figure

11. To save sdc open terminal and type the same command in the below figure
    i. saveNetlist -includePowerGround /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withPG.v

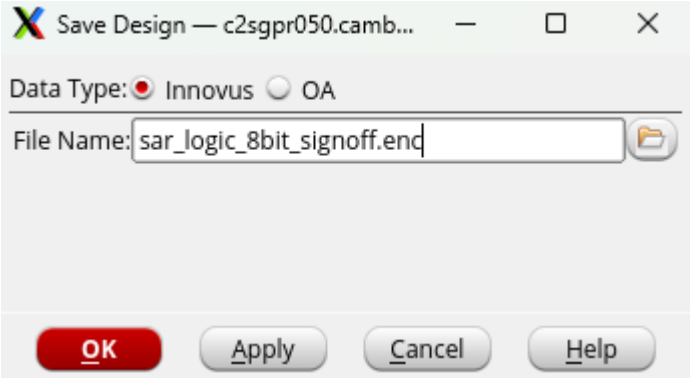
    ii.saveNetlist /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withoutPG.v

    iii. write_sdc /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit.sdc

```
innovus 11> saveNetlist -includePowerGround /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withPG.v
Writing Netlist "/home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withPG.v" ...
Pwr name (VDD).
Gnd name (VSS).
1 Pwr names and 1 Gnd names.
innovus 12>
innovus 12> saveNetlist  /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withoutPG.v
Writing Netlist "/home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit_withoutPG.v" ...
innovus 13>
innovus 13> write_sdc /home/student/CIT_ANALOG_IEP_SAR/8_CIT_SAR_SIGNOFF/./Signoff/sar_logic_8bit.sdc
```

12. Now we can save the design as signoff.enc

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13. After saving open terminal and type exit to exit innovus

14. Now PnR is completed for Physical verification we need two files for that

    i. Sar_logic_8bit.gds
    ii. sar_logic_8bit_withPG.v
    iii. sar_logic_8bit_withoutPG.v



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